Uncertainty-Driven Adaptive Frame Selection for Video Question Answering via Reinforcement Learning

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Abstract

Vision-language models (VLMs) waste compute on video by uniformly sampling frames regardless of the query. We propose a query-conditioned frame selection policy trained with PPO that treats video QA as active evidence-gathering: at each step, the policy keeps, skips, or stops watching based on accumulated evidence. The stopping criterion uses the VLM's own output entropy, requiring no additional model. We evaluate on NExT-QA, EgoSchema, and ActivityNet-QA, where we expect adaptive stopping to yield question-dependent token budgets that outperform fixed-budget baselines at equal or lower average cost.

1 Introduction

Modern VLMs trained on image and video data, such as Qwen2.5-VL, process video by sampling frames and encoding them as visual tokens. A 32-frame clip can produce upward of 10,000 tokens, dominating the context window. Existing compression methods, like attention pruning (FastV), frame merging (FrameFusion), hierarchical compression (LongVU), compress based on visual redundancy before the question is considered, and always consume a fixed token budget regardless of question difficulty. We argue that video QA is fundamentally an active evidence-gathering problem: the model should keep watching when it is uncertain, and stop as soon as it has gathered sufficient temporal evidence to answer confidently. This exploration-exploitation framing of exploring more frames vs. exploiting current evidence yields question-adaptive budgets that are both more efficient and more principled than fixed compression schemes.

2 Related Work

Static video token compression. Attention pruning (FastV), similarity merging (FrameFusion), and hierarchical compression (LongVU; VideoChat-Flash) all compress based on visual redundancy alone and apply a fixed budget regardless of query difficulty. Our method conditions both frame selection and stopping on the input query and model uncertainty.

RL for visual efficiency in VLMs. VisionThink uses RL to request higher-resolution images when needed. Quicksviewer applies RL to video cube compression. Both use fixed token budgets and neither models uncertainty over the temporal evidence accumulated so far. Our key distinction is an explicit exploration-exploitation tradeoff: the policy learns when it has seen enough, producing variable-length frame sequences driven by answer confidence rather than a predetermined budget.

3 Proposed Methodology

Problem setup. Given video $V = \{f_1, \ldots, f_N\}$ and query q, sequentially select frames to observe until the policy emits a stop action, then answer q using only the retained tokens, minimizing both errors and total frames consumed.

MDP formulation. At step t the state is the query embedding concatenated with a mean-pool of accepted frame embeddings and the current VLM output entropy H_t (computed from a partial forward pass over accepted frames so far). The action space is ternary: keep, skip, or stop. Upon stopping, the frozen VLM generates a final answer. The reward is $R = \mathbb{1}[\text{correct}] - \lambda \cdot |S|/N$, with $\lambda \in \{0.05, 0.1, 0.2\}$. Including H_t in the state directly incentivizes the policy to stop when the VLM is already confident and to keep exploring when it is not.

Selector architecture. Frame embeddings and query encodings come from the frozen Qwen2.5-VL-7B encoders. A 2-layer transformer selector takes the current state and outputs keep/skip/stop logits. We warm-start via supervised imitation of FastV scores before PPO fine-tuning with HuggingFace TRL. Entropy H_t is computed with no gradient through the frozen VLM at each step.

4 Evaluation

Benchmarks. NExT-QA (causal and temporal reasoning), EgoSchema (long-form egocentric, ~3 min), and ActivityNet-QA (open-ended activity understanding).

Metrics. Top-1 accuracy (primary); average token retention rate $|S|/N$; accuracy-at-budget curves; and distribution of stopping points per question type, showing that easy questions trigger early stopping and hard ones do not.

Baselines. Uniform sampling at 8, 16, 32 frames; FastV pruning; FrameFusion merging; VisionThink-style fixed-budget RL; full-frame uncompressed VLM as upper bound.

Interpretability analysis. We visualize stopping-point distributions across NExT-QA question types (causal, temporal, descriptive) and plot entropy H_t over time per category, testing whether the policy learns that causal questions require more temporal exploration than descriptive ones.